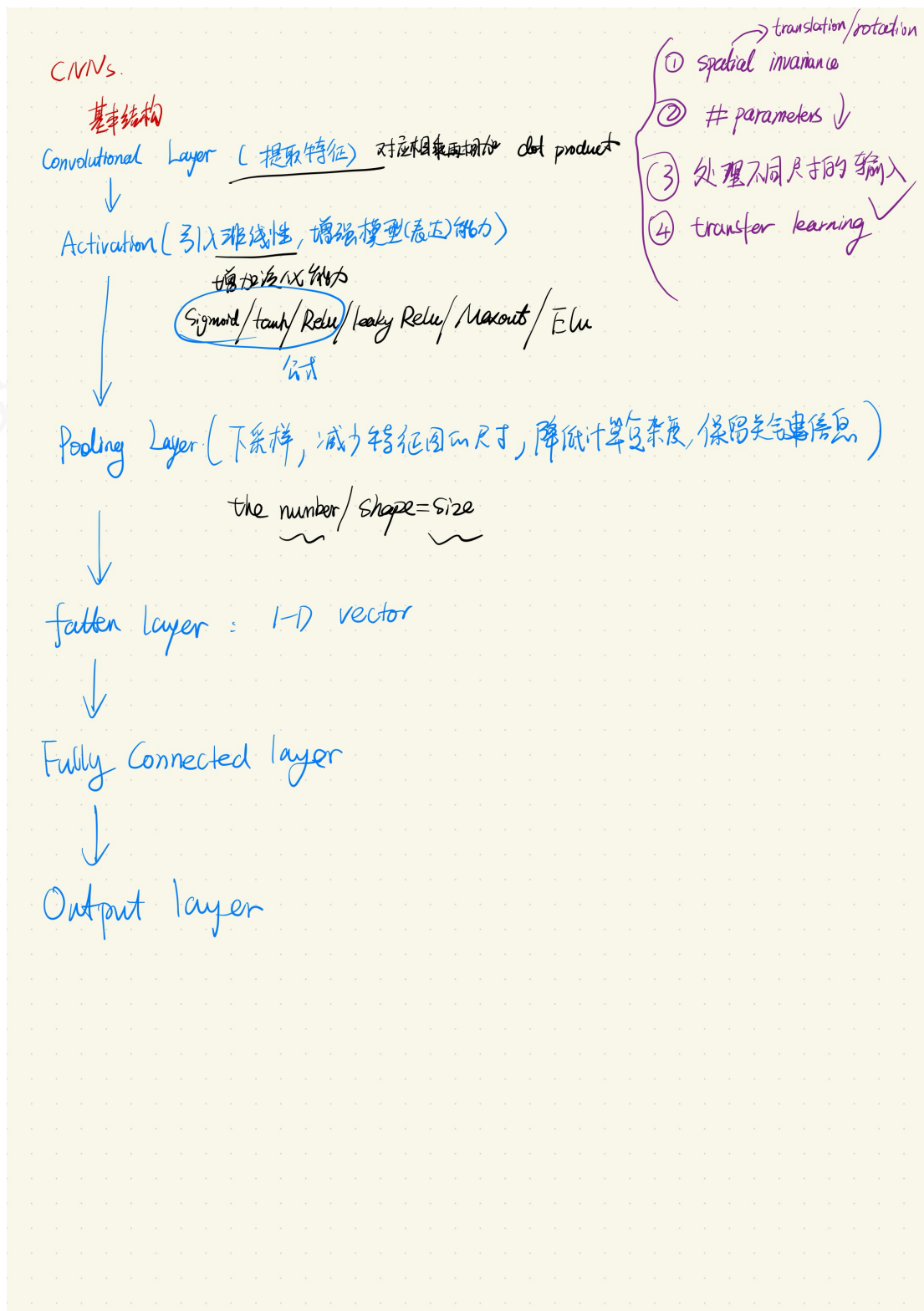






channels: 由 conv kernels 数量确定  
每个通道代表不同 kernels in feature map

size of kernel 常用 odd, 不太用 even

$$(3 \times 3) \times 3 = 7 \times 7 \begin{cases} \text{网络更深, 更加非线性} \\ \text{参数数量} \downarrow \end{cases}$$

stride:  $\text{stride/dilation} \uparrow \Rightarrow \text{size of feature map} \downarrow \Rightarrow \text{Resolution} \downarrow \Rightarrow \text{global feature}$   
 $\Rightarrow \text{receptive field} \uparrow$   
 $\text{stride/dilation} \downarrow \Rightarrow \text{size of feature map} \uparrow \Rightarrow \text{Resolution} \uparrow \Rightarrow \text{local feature}$   
 $\Rightarrow \text{receptive field} \downarrow$

padding:  $\begin{cases} \text{调整 conv 输出尺寸} \\ \text{保留边界信息} \end{cases}$

输出尺寸计算

input: image  $M$     output  $1 + (M + 2P - N) / S$   
          kernel  $N$   
          padding  $P$   
          stride  $S$

Example input image:  $256 \times 256$

step 1:  $3 \times 3$  kernel, stride 1, padding 1

step 2: Max-pooling  $2 \times 2$ , stride 2

Output size = ?

$$\text{step 1: } 1 + (256 + 2 - 3) / 1 = 256$$

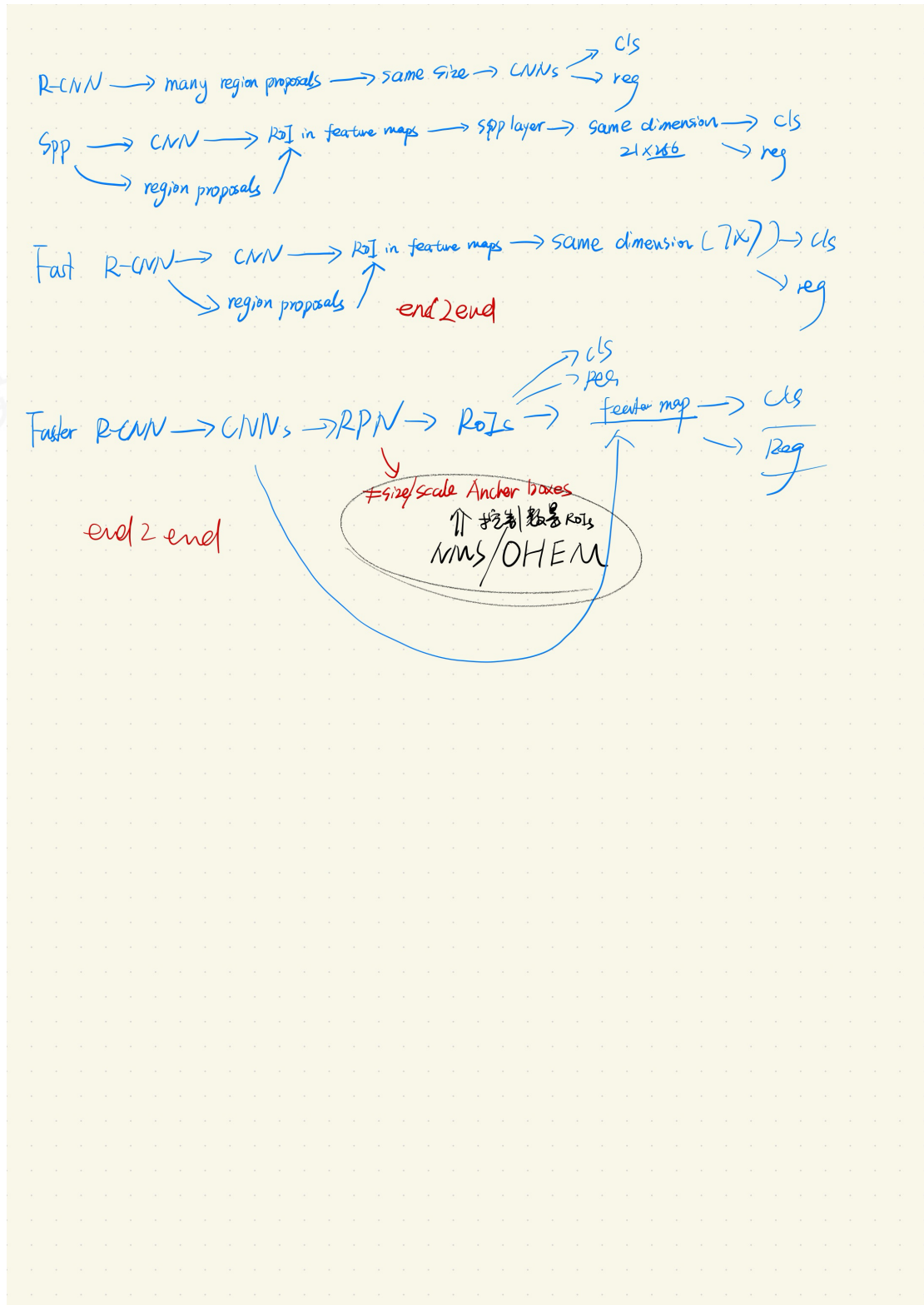
$$\text{step 2: } 1 + (256 + 0 - 2) / 2 = 1 + 254 / 2 = 128$$



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